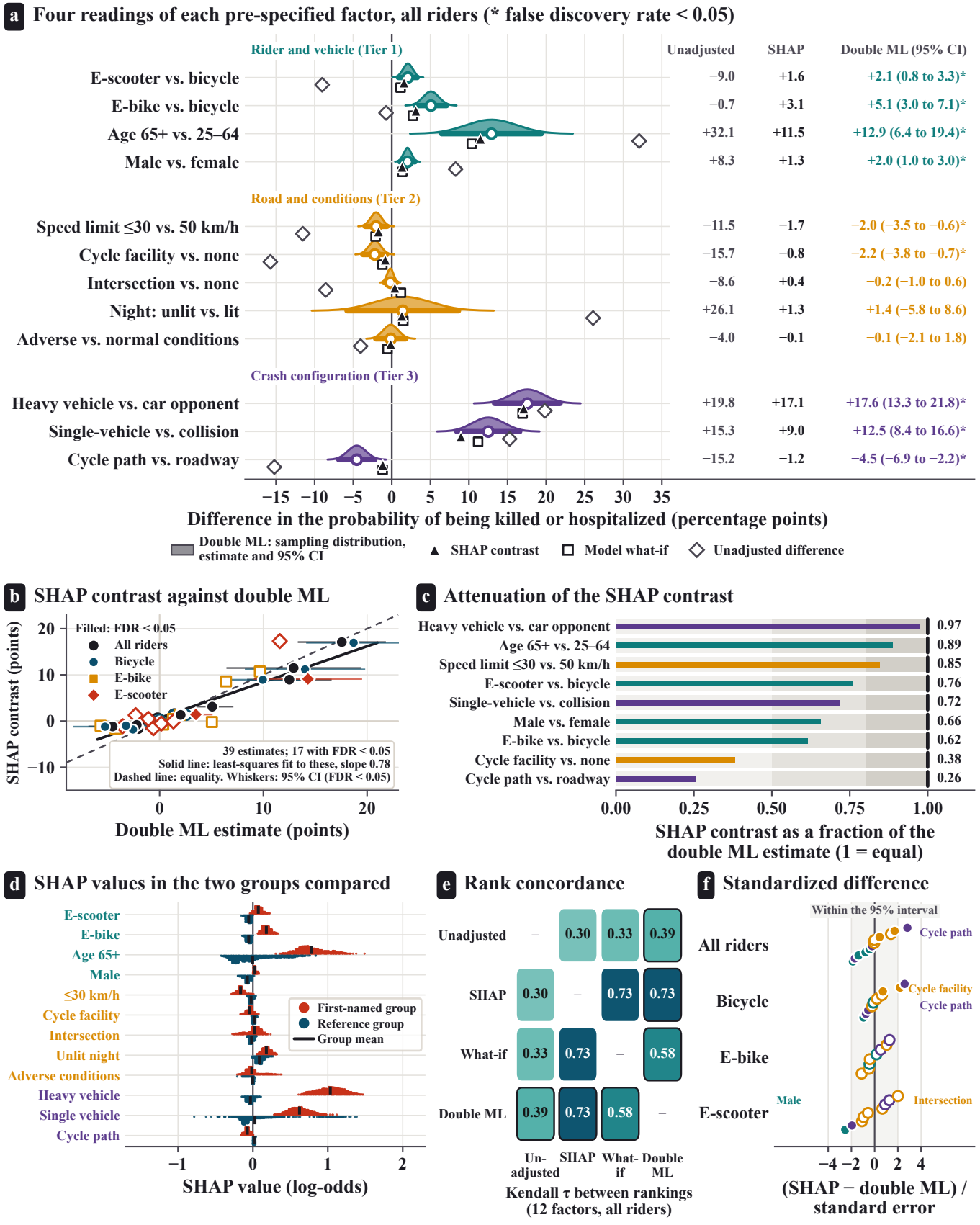

a Four readings of each pre-specified factor, all riders (* false discovery rate < 0.05)
Rider and vehicle (Tier 1)
Road and conditions (Tier 2)
Crash configuration (Tier 3)
E-scooter vs. bicycle
E-bike vs. bicycle
Age 65+ vs. 25–64
Male vs. female
Speed limit ≤30 vs. 50 km/h
Cycle facility vs. none
Intersection vs. none
Night: unlit vs. lit
Adverse vs. normal conditions
Heavy vehicle vs. car opponent
Single-vehicle vs. collision
Cycle path vs. roadway
Unadjusted
SHAP
Double ML (95% CI)
−9.0 +1.6 +2.1 (0.8 to 3.3)*
−0.7 +3.1 +5.1 (3.0 to 7.1)*
+32.1 +11.5 +12.9 (6.4 to 19.4)*
+8.3 +1.3 +2.0 (1.0 to 3.0)*
−11.5 −1.7 −2.0 (−3.5 to −0.6)*
−15.7 −0.8 −2.2 (−3.8 to −0.7)*
−8.6 +0.4 −0.2 (−1.0 to 0.6)
+26.1 +1.3 +1.4 (−5.8 to 8.6)
−4.0 −0.1 −0.1 (−2.1 to 1.8)
+19.8 +17.1 +17.6 (13.3 to 21.8)*
+15.3 +9.0 +12.5 (8.4 to 16.6)*
−15.2 −1.2 −4.5 (−6.9 to −2.2)*
−15 −10 −5 0 5 10 15 20 25 30 35
Difference in the probability of being killed or hospitalized (percentage points)
Double ML: sampling distribution, estimate and 95% CI
SHAP contrast
Model what-if
Unadjusted difference
b SHAP contrast against double ML
Filled: FDR < 0.05
All riders
Bicycle
E-bike
E-scooter
39 estimates; 17 with FDR < 0.05
Solid line: least-squares fit to these, slope 0.78
Dashed line: equality. Whiskers: 95% CI (FDR < 0.05)
SHAP contrast (points)
Double ML estimate (points)
c Attenuation of the SHAP contrast
Heavy vehicle vs. car opponent 0.97
Age 65+ vs. 25–64 0.89
Speed limit ≤30 vs. 50 km/h 0.85
E-scooter vs. bicycle 0.76
Single-vehicle vs. collision 0.72
Male vs. female 0.66
E-bike vs. bicycle 0.62
Cycle facility vs. none 0.38
Cycle path vs. roadway 0.26
0.00 0.25 0.50 0.75 1.00
SHAP contrast as a fraction of the double ML estimate (1 = equal)
d SHAP values in the two groups compared
E-scooter
E-bike
Age 65+
Male
≤30 km/h
Cycle facility
Intersection
Unlit night
Adverse conditions
Heavy vehicle
Single vehicle
Cycle path
First-named group
Reference group
Group mean
−1 0 1 2
SHAP value (log-odds)
e Rank concordance
Unadjusted – 0.30 0.33 0.39
SHAP 0.30 – 0.73 0.73
What-if 0.33 0.73 – 0.58
Double ML 0.39 0.73 0.58 –
Un-adjusted
SHAP
What-if
Double ML
Kendall τ between rankings (12 factors, all riders)
f Standardized difference
Within the 95% interval
All riders
Bicycle
E-bike
E-scooter
Cycle path
Cycle facility
Cycle path
Male
Intersection
−4 −2 0 2 4
(SHAP − double ML) / standard error